

Problem Solving Using Python (CSE1012)

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Agenda / Contents

01 List as Queues

02

03

04

05

Queues



Queues

- Queue is another important data structure which stores its elements in an ordered manner.
- Consider some analogy:
 - People standing in line to board a bus. The first person will get in first
 - Luggage kept on conveyor belt. The bag which was placed first will be the first to come out at the other end
 - Cars lined at a toll bridge. The first car to reach the bridge will be the first to leave

Queues

- In these examples, we see that the element at the first position is served first
- Same is the case with the queue data structure
- It is a First-In-First-Out (FIFO) data structure

Queues

- The element in a queue are added at **one end** and removed from the **other end**
- In computer systems, the operating system makes full use of queues for various tasks

Queues

- In computer systems, the operating system makes full use of queues for various tasks such as:
 - To maintain waiting lists for a single shared resource like printer, disk, CPU etc.
 - To transfer data asynchronously (data not necessarily received at same rate as sent) between two processes (IO buffers), e.g., pipes, file IO, and sockets
 - Handling interrupts.

Queues

- In computer systems, the operating system makes full use of queues for various tasks such as:
 - Real-time system can be interrupted, for example, by a mouse click, it is necessary to process the interrupts immediately; before proceeding with the current job
 - If interrupts have to be handled in order of arrival, then FIFO queue is the solution

Using Lists as Queue

- Queues support three basic operations:
 - insert,
 - delete, and
 - peep (or peek)

Using Lists as Queue

- In Python, we can easily implement a queue by using the `append()` method to insert an element at the end of the queue
- `pop()` method with an index 0 to delete the first element from the queue
- `Slice` operation to peep (or peek)

Using Lists as Queue

```
queue = [1,2,3,4,5,6]
print("Original queue is : ", queue)
queue.append(7)
print("Queue after insertion is : ", queue)
queue.pop(0)
print("Queue after deletion is : ", queue)
print("Value obtained after peep operation is : ", queue[(len(queue) - 1)])
```

OUTPUT

```
Original queue is :  [1, 2, 3, 4, 5, 6]
Queue after insertion is :  [1, 2, 3, 4, 5, 6, 7]
Queue after deletion is :  [2, 3, 4, 5, 6, 7]
Value obtained after peep operation is :  7
```

Thank You

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